

VR Body Calligraphy - Whole-body Performance Writing for Chinese Calligraphy

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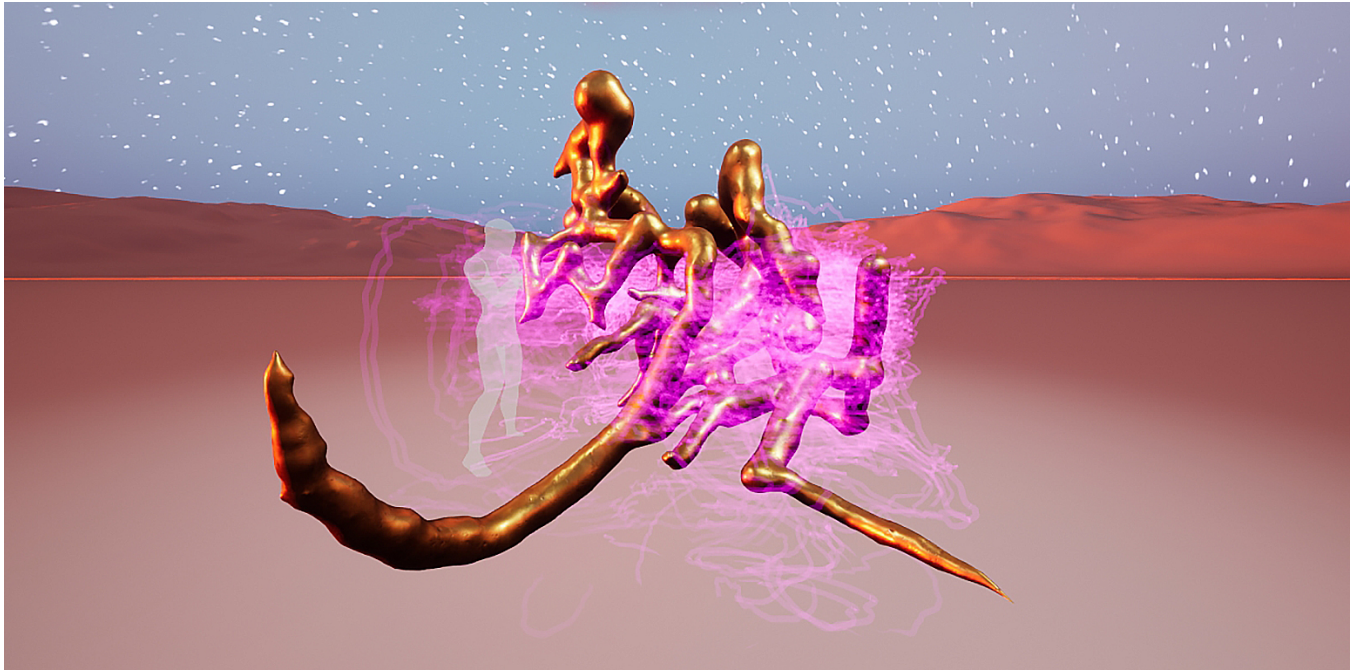


Figure 1: Prosperous Dragon, version C (Shum 2023)

Abstract

Chinese Calligraphy is a millennia-old art form. It is an intangible heritage of the Chinese culture. Despite our digital lives, this art form has largely remained unaltered in its traditional two-dimensional expression. Current research is focused on AI-driven character generation on one hand, and on the other the replication of ink and brush dynamics within digital environments, as 2D simulations of the traditional paper canvas. Instead of imitating the existing, this research articulates a technical, artistic, and cultural evolution of Chinese Calligraphy Shufa 书法 from the traditional two-dimensional, material-based expression to a novel digital three-dimensional form within virtual reality. The paper is

structured as a series of cross-disciplinary experiments amalgamating the essence of the traditional craft with today's possibilities in computer-human-interactions. Each is evaluated and discussed with an established traditional calligraphy master Law Ching-Bor, commenting on the transformation from single character to complex compositions. At the same time, advancements in finger and full body tracking technologies facilitate sculptural inscription in virtual space. We attempt to write calligraphy with choreographed whole-body gestures, translating these movements into dynamic, three-dimensional strokes of digital Shufa. The investigation details the scribing, interpreting, and experiencing multi-character calligraphy in a volumetric VR context. Ultimately, this paper presents Chinese calligraphy not as a relic of the past but as a living tradition—morphing, expanding, and resonating within the digital context, thereby transforming and evolving its cultural legacy.

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CCS Concepts

• Applied computing → Performing arts; Media arts; • Human-centered computing → Virtual reality.

Keywords

Culture, Craft, Drawing, Human interaction, Immersion, Multimedia, Performance, Sculpture, Virtual Reality, Visualization

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1 Introduction

Traditional Chinese Calligraphy, *Shufa* 书法, articulates the physical writing of ink on paper using brushes. It is a deeply material-based practice that also transcends beyond art, to a core philosophy of Chinese culture, recognized by UNESCO (2009). It is a tangible practice with an intangible cultural heritage - a lineage spanning millennia. Our research aims to extend this practice further, from a material-based art into a digital form of media dissemination. Retaining the core values of *Shufa* 书法, these are referred to as *geist* (spirit) in the West and represented by *yi* (意, intent) in Chinese [Shi 2019]. Instead of using ink and brush, the physical tools articulating a two-dimensional practice, we investigate how three-dimensional calligraphy can be written in VR space. This paper will present our latest findings in engaging a full-body scale performed calligraphy writing multiple characters, and contextualise them with traditional *Shufa* through discussion and comparison with Master Law Ching-Bor 罗澄波. Master Law is an internationally recognized Hong Kong based master calligrapher, who was twice selected (1995 and 1999) for exhibition in the China National Calligraphy and Seal Inscription Exhibition, one of the highest tiers of recognition in this field. His masterpiece "You Yuanmingyuan Guzhi" (Visit to the Old Summer Palace, 1999) was inscribed on the China Changde Poem Wall as a national monumental historical relic.

This paper is structured in the following parts. First, we will introduce the background of our research, and summarise the previous results of using a hand-tracking system to trigger digital ink within VR. In the second part, we investigate writing multiple-character works in VR space, using full-body capture with VR calligraphy in the context of performance and choreography. In the last part, we discuss our most recent multi-character work *Prosperous Dragon* against the discourse of traditional calligraphy. Having viewed these works, Master Law discusses his thoughts throughout the process to provide important feedback and evaluation of the research.

2 Background

Our exploration of Chinese calligraphy begins with examining the relationship between technology and Chinese philosophy [Hui 2016], establishing a conceptual framework in the pursuit of *The Spirit of Chinese Art*. The intent is "to evolve the tools and media of calligraphy, melding its material-based phenomenology together with its philosophical duality to retain its artistry. At the same time, we aim to transfer its qualities to an immaterial and dynamically time-based version in VR" [Shum and Klein 2023]. Previously [ibid], we have started unravelling the philosophical aspects of the research, and written in depth about the aesthetic considerations of transforming two-dimensional calligraphy into three-dimensional

forms. This paper aims to specifically focus on exploring the complexities and cultural evaluations of writing calligraphy via choreograph and performance-like movements. It reflects on how this new approach affects the visual representation of the calligraphy in the context of traditional *Shufa*.

New paradigms of calligraphy made possible by new media and tools

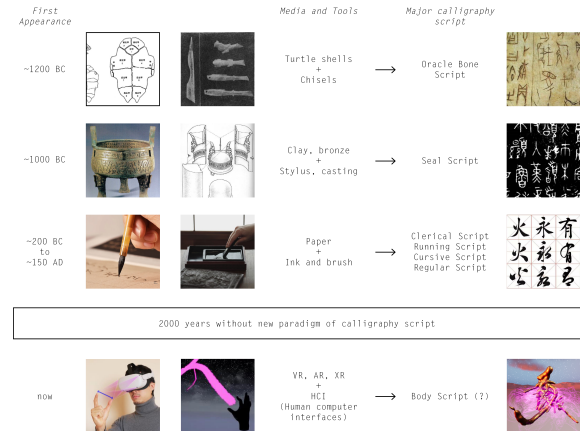


Figure 2: New paradigms of calligraphy made possible by new media and tools

This is in many ways not new, as the evolution of *Shufa* is closely linked to the development of tools and the medium of writing itself (fig2). This development extends from the chiseling of turtle shells with knives in Oracle Bone Script (c. 1200 BCE) [Qiu et al. 2000] to the standardized use of ink and brush on paper [Chang 1938]. The current age of VR, AR, and XR spaces offers a radical new medium and, with it, a new format for the development of calligraphy in the extended dimensionality of these spaces. This new medium requires a departure from the brush and ink, to a digital materiality. Thus, there is a need and opportunity to create a new tool set, one that is appropriate for this emerging medium with this evolved materiality [Boden and Edmonds 2019]. In our testing of existing VR drawing and sculpting apps such as Tilt Brush, Medium, Gravity Sketch, and Kodon, we realised that these apps only allow the writing of basic, string like calligraphy. Due to the limitation of using a generic VR controller, it was not possible to write with a precise and real-time changing of stroke width, with a dynamic ink-like writing effect [Shum and Klein 2023]. The closest VR Chinese calligraphy app available to simulate this important material behaviour is "You, Calligrapher", where a reductive and basic emulation of writing Chinese calligraphy on flat paper can be experienced. To our knowledge, there is currently no VR app that allows for writing fluid and three dimensional Chinese calligraphy. Likewise, there is no research on this topic of sculpturally expressive Chinese calligraphy, hence the need for us to develop our experience first hand.

We developed a tool-set that is built to "retain tradition yet extend new possibilities within multidimensional space and synergy of new virtual materiality", using the existing hardware of an optical

hand-tracking system with customized software development to trigger digital ink that could suspend in 3D space inside VR [ibid].



Figure 3: Left: Tiger (Law, 2022); Right: Tiger, Front and Top View (Shum, 2022)

The result of this development, *Tiger* (fig 3), was written in one set of fluid and continuous full-body motions, leading to a three-dimensional and sculptural calligraphy [Shum and Klein 2023]. The aim to express a dynamic flow of ink with an upright sculptural balance was achieved, and is not dissimilar to the qualities of traditional calligraphy. We note though that the act of writing a well-composed character in 3D space whilst coordinating the hands, body, and feet in tandem is challenging. This work required 88 takes across three days, as a precise set of choreographed movements [Leon 2022] was required to write smoothly and precisely in 3D at body scale. As we strived to retain the same focus of writing within the linearity of time in calligraphy, *Tiger* was created without the use of *undo*. Hence, as per tradition, errors would result in restarting from scratch, adding value to the writing itself, through the irreversible act thereof.

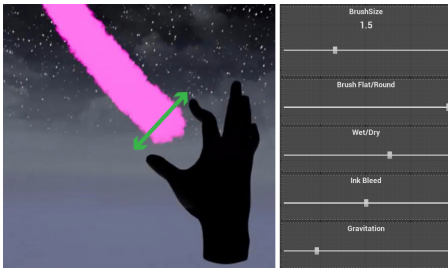


Figure 4: Left: Distance between fingers is tracked in real-time to control the width of the calligraphy stroke. Right: Various aesthetic ink and brush parameters that can be altered. (Shum 2023)

Technically, our former setup is based on the Quest 2 VR system. This system allows for accurate optical tracking of the hands and fingers (fig4) via the cameras on-board the head mounted display (HMD), and no controllers are utilized. Our programming is coded in the Unreal Engine platform, with the setup of such documented previously [Shum and Klein 2023], and now combined with Sony Mocopi as motion capture hardware.

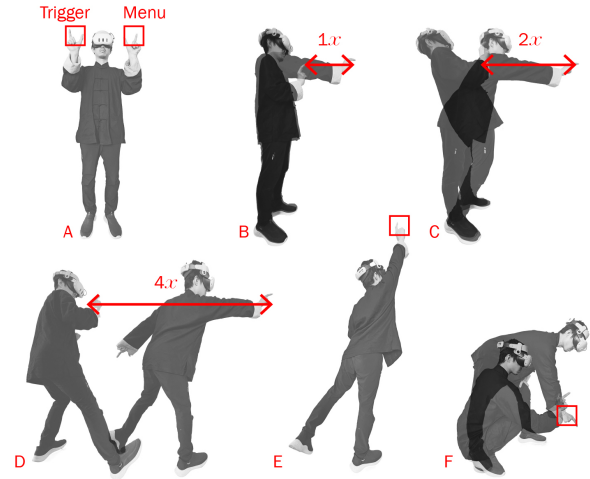


Figure 5: Body pose considerations during calligraphy writing by the author (Shum 2024)

3 Expanded Body Movements

In the writing of *Tiger*, looking in particular at the top view (fig 3), we could see the similarities between our work and that of dance choreographers embodying three dimensions instead of the two-dimensionality inhabited by traditional calligraphy. In particular, *Synchronous Objects for One Flat Thing, Reproduced* [Maria Palazzi and Forsythe 2011] demonstrates how complex dance choreographies can be systematically broken down "to enrich cross-disciplinary investigation and creativity by revealing deep structures of choreographic thinking." [ibid]

Our method of writing calligraphy creates three-dimensional calligraphy using a full range of body motion. In order to compare our practice with contemporary choreography, we shall expand on the full range of body movements required in writing such calligraphy.

At the start of writing, we always take the base pose (fig 5A), where both hands are raised for the hand-tracking (tracked from the camera on the headset) to be enabled. The left hand has two functions, triggering ON/OFF states of the right hand - the main writing hand. In the ON state, the right hand writes calligraphy by generating ink-like particles. Secondly, the left hand can trigger the menu, where we can restart, save, or adjust various parameters of brush thickness, ink flow rate and ink bleed (fig4). As mentioned, there is no UNDO functionality as we believe this breaks the flow of creation, and is against the spirit of writing calligraphy. When the right hand's state is ON, the distance of between the thumb and index fingers is tracked in real-time. Smaller distances result in a thinner stroke, and larger distances - a thicker stroke.

In our calligraphy, as we move from a 2D (height and width) to 3D (height, width and depth) expressions of calligraphy, we need to focus on how to create added depth intentionally. As figure 5B shows, with the body static, the depth available is only an arm's length, and arching the spine forwards or backwards (fig 5C) can only increase the depth slightly. Therefore, the combination of

proper footwork with the intricate co-ordination of the writing hand (fig 5D) is essential to the realization of depth in our practice.

As we start to write larger and multi-character calligraphy, the range of movement increases. In some instances (fig 5E), we need to reach the highest possible height, but simultaneously maintaining control (without jumping), and pre-planning the footwork required in the next steps. Similarly, when writing at a low height (fig 5F), there are the options of bending the back downwards or squatting down. The former allows the feet to still move easily, but is very tiring when writing for a long time, and the opposite applies to squatting down.

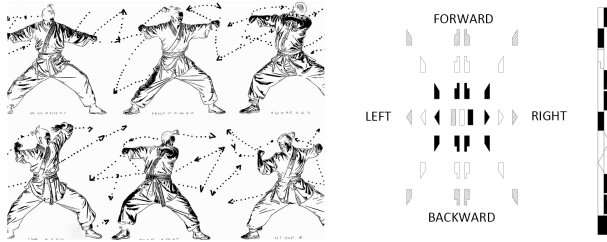


Figure 6: Left, Diagram showing forces of motion in Kungfu (Shum, 2024). Right, Interpretation of Laban Kinetography footwork (Shum, 2024)

The above only meticulously analyses a fraction of the body movement considerations we have to account for when writing our calligraphy. Thus moving on, as we continue to document more movements in ever more complex character arrangements, we might need to consider creating a catalogue of choreographed characters, to systematically notate the body movements involved in creating this novel typography of Chinese calligraphy. This could perhaps be done by referencing the diagrams of Kungfu [Li et al. 2006] and dance notation, such as Laban Kinetography [Karl-Johnson 2017] (fig6).

3.1 Initial implementations of body tracking and multi-character calligraphy in VR



Figure 7: Dragon with Movement Traces, Front View, Top View, and Side View (Shum 2023)

We now present our calligraphy works from a single character *Dragon*, then to a work of four characters *Prosperous Dragon*. The intention is to investigate how writing such calligraphy in VR via recorded body movements could enrich or juxtapose with Chinese

calligraphy. We invited Master Law to view both works first on a flat display and then within VR, both as final written forms and as animated replays. His reflections offer us an important perspective to validate and further improve our research.

Our work *Dragon* (fig 7) is the result of adding an embodied avatar via live full-body motion capture. Additionally, we visualized the *trails* from the choreographed movements of the writer's body to become part of the calligraphy. (Video [here](#), password 123123.) The work is written in three motions as illustrated in figure 7, along with the corresponding footwork (L for left foot and R for right foot). With the recorded avatars, we can articulate the scale relationship between the work and the calligrapher. This allows the authors and viewers to have an enriched understanding of the underlying body coordination, fluid movement, and writing speed required for the creation of the works, and opens a possibility for discourse with traditional calligraphy, where the speed and agility of the brushwork is part of the finesse of writing.

With our implementation of movement traces, the choreography can be envisaged as a visual object [Leach 2014] of the entire body synthesized with the writing process. We believe for the first time in Chinese calligraphy discourse, we can amalgamate both the calligraphy with its writing process in the same *space-time* and within the same *media*. For us, the semi-transparent tracing of movement manifests the memory of our writing as a piece of digital material, where one whole movement over time can now be condensed into a static, yet visually dynamic cloud of movement. This pairing of the calligraphy with the movement traces reveals a new perspective of understanding of our works. We are now able to determine where hand extensions can be adjusted to give the calligraphy added sculptural depth, and where footwork can be corrected to achieve better writing fluidity. This allows us to refine our training to write improved calligraphy.

Master Law's thoughts, on scale: Traditionally calligraphy writing is limited to the media including the sizes of brush, paper, and writing surface, and is predominantly written with the hand and arm. In VR space, our approach is free from such physical constraints and allows for an added range of motion, writing instead with the whole body. Master Law notes the difficulty of appreciating the "wholeness" of the works on a flat screen, in comparison with the "monumental and sculptural" quality when viewing in VR. The fact that the calligraphy is created via whole-body movements is only evident when experienced at true scale inside VR. Master Law remarks that the scale, immersion, and complexity of the works in VR encourage him to move around, explore, investigate, and "inhabit" the works, which is not possible at all with traditional calligraphy.

On the body: Master Law appreciates the visualization of the body with the traces of movement. He notes that the human body now becomes a "second brush", and more care ought to be given to the visualization of the movement traces. He notes that perhaps a more selective and cleaner approach to the visualized trails would give greater clarity compared to the current "cloud-like" effect. This way, the writer has to give even greater thought to the choreography of the body, almost like dancing with a partner in sync. He reminds us that in calligraphy, it is often said the calligrapher is dancing with the brush.



Figure 8: Prosperous Dragon, version A. Left: Front view. Right: With movement traces, side view, top view (Shum 2023)

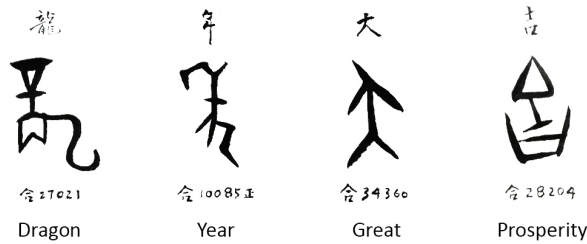


Figure 9: Oracle Bone Script references for *Prosperous Dragon* (Law 2023), referencing [Liu and Feng 2019]

The next work *Prosperous Year of the Dragon* (referred to as *Prosperous Dragon* from now), consists of the four characters of Dragon 龍 Year 年 Prosperous 大吉. *Prosperous Dragon* is conceptually based on Oracle Bone Script references [Liu and Feng 2019], which is the earliest form of Chinese writing [Qiu et al. 2000]. Master Law wrote these characters in this script to serve as references (fig 9). It is also our first work using our newly developed body-tracking set-up for a multi-character work.

In Version A (fig 8, video [here](#), password 123123), we tested writing the four characters, arranged in a cube-like formation, each character touches the next to form a single composite and sculptural object. Writing them from the outside, choreographing them inwards emphasised the sculptural quality above omni-directional readability. However, from all viewing angles, the characters are partially occluded from the other characters, decreasing legibility. Yet, in the animated replays of the recorded writing process, we could easily recognise characters as they are being written out. This creates a conundrum between a legible writing process versus an abstract sculptural form.

Master Law, on form: Traditional calligraphy discourse has always viewed and perceived the final calligraphy as the validated work. This certainly is due to a lack of tools to record the creative process in its history over three millennia. Even within contemporary calligraphy [Barras 2002], there is no precedent for combining the process and the calligraphy within one medium and singular

space-time, as this research demonstrates. In *Prosperous Tiger*, we observed the simultaneity of the calligraphy writing process, collapsing with the final form within the same, digitally simulated materiality. This poses a conundrum in calligraphy as the version that is legible during its writing process seemingly is not as valid as one that is only legible in its final form. Master Law views the performative aspect of the research as having equal weight to the final written form of the works.

4 Prosperous Dragon - Critical Evaluation of VR Body Calligraphy in the context of traditional Chinese Calligraphy - with Master Law Ching-Bor

The final work, *Prosperous Dragon* (fig 10) is a complex arrangement of the four characters based on the discussions with Law, written in an abstract sculptural composition, barely legible from any direction due to occlusion of the characters. Compared to the rigid box-like arrangement of the initial Version A (fig 8), we have refined the sculptural quality of the work by having the four characters intertwine and connect at different heights and angles, reacting to Law's comments on form.

This work is related to the extreme variations of the Cursive script, the Wild Cursive. Calligraphers such as Zhang Xu (656-713) and Huai Su (737-799) wrote with such freedom and energy that characters would require study to be deciphered. This practice was pushed further by contemporary calligrapher Wang Dongling in his Chaos Script works, where no characters are seemingly legible, and the brushstrokes strive only to reflect the spiritual and emotional qualities of calligraphic expression [Iezzi 2015]. This juxtaposition of *legibility with time against illegible sculptural form*, transferred to our time-based version of calligraphy could be considered an evolution in the calligraphic discourse.

Master Law, on legibility versus abstraction: In our experiments, with the increased dimensional perception and issues of viewing angle, we thus attempted to write characters that can be legible. Even so, characters might also be illegible by occlusion, as Version A demonstrates. Master Law notes increased legibility during the animated replay, but as a calligrapher with over 60 years of experience, he has no difficulty understanding the final written forms, and appreciates the sculptural quality above simple readability. For him, the notion of legibility has always existed in the calligraphy discourse and should not be dominant in this new form. He advises though, in the spirit of exploring a radical evolution of the medium of calligraphy, to maintain a degree of legibility to disseminate Chinese calligraphy and not to digress into the development of scripts similar to the Wild or Chaos scripts for now.

Within this novel and immersive environment to appreciate calligraphy, there are many extensions to our perception [Berger 1972; Grau 2003; Hoppe et al. 2022]. Traditional calligraphy is flat with only one frontal perspective to view it. However in our works, and there is no defined "front viewing angle", apart from the angle where a character can be legible.

On perspective: To resolve the viewing angle issue, Master Law suggests a way of "framing the canvas". A piece of traditional calligraphy has definite bounds in the size of the paper, with the signature



Figure 10: Prosperous Dragon, Top left: Particle effect version as displayed during an animated replay. Top right: Voxelized version as final written form. Bottom: side and top views. (Shum 2023)

and red seal added to enhance the framing effect. Likewise, such items could be added as "evolved versions" within 3D space to give bounds and directionality to the works in VR.

5 Conclusion

The overall aim of our research is to evolve Chinese calligraphy in a new medium whilst maintaining the *Spirit of Chinese calligraphy*. Master Law defines the basic act of calligraphy as the art of writing Chinese characters with expressive lines. As a conclusion, he commends the current research direction which satisfies this criterion by the dynamism and fluidity of the written strokes shown in the calligraphy works. Taking the advice from Master Law, we have identified two important future outlooks in regards to this research.

We have created a novel tool that enables a new format of sculptural and choreographed Chinese calligraphy writing, which can be recorded along with the creator's body in motion as an avatar via full-body tracking. With this novel tool, a set of questions has emerged. How much of the body movement with its traces, should be considered part of the calligraphy? This raises a conundrum regarding the relationship between the traces and the calligraphy itself and warrants further study.

Second, we have begun investigating issues of spatial composition and legibility in multiple-character calligraphy, paving the way

for more refined and complex works that engage in dialogue with traditional calligraphy. However, as the complexity of our work increases, the need for a clear catalogue of whole-body movements has arisen. Therefore, we will consider creating a catalogue related to the whole-body movements involved in writing this calligraphy, serving as a novel spatial archive or typography of Chinese calligraphy, and continue in our experimentation to present Chinese calligraphy as a contemporary continuation and extension of its cultural heritage.

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